

SARYG-SEP, Russian Federation

WMO: 361040

Lat: 51.4789N

Lon: 95.5778E

Elev: 706

StdP: 93.13

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -42.3	(c) -39.9	(d) -44.9	(e) 0.0	(f) -42.4	(g) -42.5	(h) 0.1	(i) -39.9	(j) 3.6	(k) -24.8	(l) 3.4	(m) -24.8	(n) 0.2	(o) 120	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.0	(c) 32.2	(d) 17.7	(e) 29.7	(f) 16.9	(g) 27.7	(h) 16.4	(i) 19.3	(j) 28.0	(k) 18.5	(l) 26.4	(m) 17.7	(n) 25.1	(o) 2.1	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	13.0	21.1	15.9	12.3	20.3	15.1	11.7	19.9	58.3	27.6	55.5	26.6	53.0	25.3	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.3	(b) 4.3	(c) 3.7	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(4) 5.3	(5) 4.3	(6) 3.7	(7) -44.2	(8) 20.8	(9) 2.0	(10) 1.0	(11) -45.6	(12) 21.5	(13) -46.8	(14) 22.1	(15) -47.9	(16) 22.6	(17) -49.3	(18) 23.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.6	-29.7	-24.1	-8.9	4.6	9.9	17.4	19.4	17.3	10.3	1.5	-13.2	-24.9		
	DBStd	17.87	5.04	6.14	7.70	4.47	4.56	3.85	2.79	3.22	4.31	4.77	7.42	5.89		
	HDD10.0	5091	1230	955	587	166	59	1	0	0	49	266	697	1082		
	HDD18.3	7384	1488	1188	845	411	262	61	19	58	242	522	947	1340		
	CDD10.0	864	0	0	0	6	57	225	292	226	57	2	0	0		
	CDD18.3	114	0	0	0	0	1	34	53	26	1	0	0	0		
	CDH23.3	1872	0	0	0	4	105	645	711	364	42	0	0	0		
	CDH26.7	658	0	0	0	0	25	245	256	126	6	0	0	0		
(14) Wind	WSAvg	1.7	1.1	1.5	1.8	2.1	2.3	2.0	2.0	1.8	1.9	1.7	1.4	1.2		
(15) Precipitation	PrecAvg	333	9	6	8	16	26	47	73	64	36	17	18	14		
	PrecMax	430	19	15	17	29	42	67	127	104	63	38	36	26		
	PrecMin	225	4	1	2	4	12	25	32	25	15	5	8	5		
	PrecStd	51	4	3	4	6	9	12	26	22	14	8	7	5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-12.2	-4.9	10.6	23.8	30.5	34.2	35.6	33.5	27.4	19.0	5.8	-5.4		
		MCWB	-12.3	-5.7	3.6	10.7	14.1	18.0	19.0	18.5	15.0	8.5	0.9	-6.0		
	2%	DB	-16.8	-9.6	7.3	20.0	26.7	31.7	32.3	29.9	24.4	14.8	1.8	-10.7		
		MCWB	-17.2	-10.4	2.3	9.4	13.3	17.2	18.4	17.7	14.3	7.3	-0.5	-11.3		
	5%	DB	-19.1	-11.7	4.6	16.6	23.7	29.5	29.3	27.1	21.4	11.4	-0.8	-13.7		
		MCWB	-19.5	-12.5	0.8	7.4	11.8	16.6	17.4	17.2	13.0	5.7	-1.9	-14.2		
	10%	DB	-21.4	-13.8	1.9	13.9	21.0	27.3	27.3	24.6	18.3	8.6	-2.8	-16.4		
		MCWB	-21.8	-14.5	-0.6	6.0	10.8	15.9	17.0	16.2	11.6	4.4	-3.8	-16.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-12.4	-5.9	4.3	11.8	16.1	20.2	20.3	20.4	16.1	9.7	1.7	-6.1		
		MCDB	-12.1	-4.7	9.7	20.9	27.1	30.4	30.5	29.8	24.3	17.7	4.3	-5.1		
	2%	WB	-17.0	-10.3	2.5	9.7	14.0	19.0	19.3	19.0	15.0	7.8	-0.4	-11.2		
		MCDB	-16.8	-9.6	6.7	18.2	23.9	28.4	27.6	27.0	22.7	13.6	1.4	-10.6		
	5%	WB	-19.4	-12.5	1.0	7.8	12.7	18.0	18.6	18.0	13.8	6.1	-1.9	-14.2		
		MCDB	-19.0	-11.7	4.0	15.3	22.0	26.2	26.2	25.2	19.9	11.0	-0.7	-13.8		
	10%	WB	-21.7	-14.5	-0.6	6.4	11.1	16.9	17.9	17.0	12.1	4.5	-3.8	-16.7		
		MCDB	-21.3	-13.9	2.0	13.3	19.7	25.1	25.3	23.1	17.5	8.0	-2.8	-16.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	13.8	14.1	14.1	15.1	15.7	14.0	13.5	14.5	11.8	10.1	8.9		
		MCDBR	11.0	13.4	12.6	19.0	21.2	20.4	18.4	18.4	19.3	16.6	10.5	8.9		
		MCWBR	10.6	12.6	8.0	10.1	10.3	8.3	7.0	8.1	9.3	9.4	8.1	8.5		
	5% WB	MCDBR	11.2	13.5	12.2	16.7	18.8	17.4	14.7	15.3	17.3	16.0	10.1	8.8		
		MCWBR	10.8	12.7	7.8	9.6	9.8	8.0	6.3	7.5	9.0	9.3	7.9	8.5		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.296	0.343	0.386	0.407	0.393	0.396	0.380	0.336	0.320	0.299	0.271		
	taud		2.128	2.081	2.056	2.195	2.176	2.326	2.349	2.362	2.432	2.376	2.215	2.097		
	Ebn at Noon		734	809	836	848	846	862	853	848	853	788	684	646		
	Edh at Noon		74	104	131	130	139	121	116	109	90	78	69	65		
(44) All-Sky Solar Radiation	RadAvg		1.37	2.52	4.20	5.07	5.79	6.38	5.86	5.04	3.88	2.41	1.31	1.01		
	RadStd		0.05	0.09	0.14	0.25	0.34	0.44	0.34	0.29	0.35	0.16	0.10	0.04		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page